

Exam 1 Cheat Sheet

Data Types: Quantitative = numeric (Discrete = countable; Continuous = measured/any value).

Qualitative = categories (Categorical/Nominal = no order; Ordinal = ordered, e.g. grades).

Parameter = describes population. **Statistic** = describes sample. **Measurement Levels:**

Nominal (categories) → Ordinal (ordered) → Interval (ordered, equal diffs, no true zero — temp °F) → Ratio (true zero — weight, time).

Experiment = researcher assigns treatment. **Observational** = just observes (Cross-sectional = one time point; Case-control = compare w/ vs w/o condition, looking back; Cohort = follow forward in time). **Randomized experiment** = researcher (not subject) randomly assigns treatment. **Matched pairs** = same variable measured on same/matched objects. **Blind** = subject doesn't know treatment. **Double-blind** = neither subject nor researcher knows. **Neither** = both know (e.g. subject picks).

Sampling bias = sample not representative (unequal selection chances). **Stratified sampling** = divide pop. into groups, random sample proportionally from each. **Sampling with replacement** = selected item returned to pool. **Practical significance** = real-world meaningful result regardless of stats.

Probability

- Sample space = all outcomes. Event = subset of outcomes. Impossible event → $P=0$.
- **Addition Rule:** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- **Mult. Rule (independent):** $P(A \text{ and } B) = P(A) \cdot P(B)$
- **Complement:** $P(\text{not } A) = 1 - P(A)$
- **Conditional:** $P(A|B) = P(A \text{ and } B) \div P(B)$
- **Empirical:** $P(\text{event}) = \text{count of event} \div \text{total trials}$

Random Variables

- Random variable (X) = unknown quantity depending on chance.
- Discrete RV = listable possible values.
- Probability distribution = table of values & probabilities ($\sum P = 1$).
- Probability histogram = graph of values (x-axis) vs probabilities (y-axis).
- $P(a < X < b)$ = sum of probabilities strictly between; $P(X \geq a) = 1 - P(X < a)$.

Binomial

Requirements: fixed n trials, 2 outcomes, constant p, independent trials. $P(X=k) = C(n,k) \cdot p^k \cdot (1-p)^{n-k}$, $C(n,k) = n!/[k!(n-k)!]$

Graphs — Which to Use

Data	Graph
1 categorical var	Bar chart / Pie chart
1 quantitative var	Histogram / Stem-leaf / Dot plot
2 quantitative vars	Scatterplot
2 categorical vars	Contingency table

- **Relative Frequency** = frequency ÷ total (should sum to 1).
- **Contingency table**: rows × columns of categories; conditional prob = cell ÷ row or column total.
- **Stem-and-leaf**: stem = all but last digit; leaf = last digit, sorted least→greatest, comma-separated; no data = "DNE". Key shown as e.g. 12|3 = 123.

Quick Decision Tree

1. Researcher assigns treatment? Yes → Experiment / No → Observational
2. Experiment: who's unaware of treatment? Neither subject/researcher → Double-blind; only subject → Blind; both aware → Neither
3. Observational: one time point → Cross-sectional; compare condition groups backward → Case-control; follow forward → Cohort